



Ogilvy

DESCRIPTIVE ANALYTICS REPORT

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Introduction to Business Analytics

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Descriptive Analytics Report

Executive summary:

As the Marketing Campaign Manager at Ogilvy International, this report uses Descriptive Analytics on a worldwide marketing dataset to support evidence-based choices across countries, languages, segments, and demographics. The analysis has three goals: (1) rigorous data preprocessing to ensure analytical accuracy, (2) using descriptive techniques and visualisations to investigate key performance relationships, and (3) interpreting findings to inform strategic decisions for Ogilvy's international operations.

Data was first cleaned and standardized by eliminating structural errors, converting variables to appropriate formats, and resolving a tiny fraction of missing data with Predictive Mean Matching and simple imputation following an MCAR-consistent pattern. Outlier tests revealed that no extreme values required modification, establishing a solid foundation for investigation. Five scenarios were created to investigate performance by customer segment, language-location, age group and language, language-gender, and the interplay between segment and language.

Across these scenarios, two key insights emerge. First, segment performance is quite uneven: Outdoor Adventurers provide the best mix of high conversion, high ROI, and cheap acquisition cost, whereas Fashionistas have poor conversion, low ROI, and high acquisition cost, suggesting a saturated and costly category. Second, language and demography are important: German- and Mandarin-language advertising routinely beat English, particularly among Generation Z and German women, but some German and Mandarin segments exhibit mismatch between engagement and conversion. As a result, the paper proposes reallocating investment to high-performing segment-language clusters with localized creatives, as well as revamping tactics for saturated fashion and research-intensive tech audiences in order to convert engagement into long-term performance.

Introduction

1.1. Business Background

As Ogilvy expands across diverse regions, languages, and media channels, understanding campaign performance has become increasingly complex. Managing enormous amounts of global marketing data without a structured analytical methodology increases the risk of inconsistent judgments, operational inefficiencies, and lost optimization possibilities. To address this, the report uses Descriptive Analytics, which summarizes past performance to explain what happened (Davenport & Harris 2017; Shmueli & Koppius 2011).

Analytics is critical in digital marketing for assessing performance, tracking engagement, targeting consumers, and optimizing advertising campaigns. Descriptive Analytics achieves this through statistical summaries, aggregation, and visualisation, enabling pattern detection, market comparison, and clear communication of insights. Its key characteristics - data reduction, pattern identification, and high interpretability - support both technical and non-technical stakeholders (Chen et al. 2012).

For Ogilvy, these techniques improve visibility of market differences, channel and segment performance, and resource allocation, while strengthening coordination across international teams. By converting complex datasets into actionable insights, Descriptive Analytics supports evidence-based decision-making across global campaigns. The overall goal of this study is to assess how well Ogilvy's digital marketing initiatives achieve important business results, allowing for better informed budgeting, strategic modifications, and future planning. This review also determines which tactics provide the highest ROI and engagement, ensuring that the organization is prepared to make data-driven enhancements to campaign performance.

1.2. Objectives

For the benefits described above, this study intends to serve three major objectives:

1. Provide extensive data preparation to ensure analytical correctness.

2. Use descriptive analytics approaches to investigate critical performance correlations.
3. Interpret results to help Ogilvy make educated, data-driven decisions about its worldwide operations.

2. Methodology & Data Pre-Processing

2.1. Error Identification

- Identify inconsistencies

The data audit revealed some structural discrepancies that needed to be corrected. Three essential improvements were made to assure data quality.

To begin, all categorical variables were originally kept as character strings, which were then turned into factors to allow for correct grouping, summarization, and visualization. Second, the Duration variable comprised text entries - "60 days", which made analysis difficult; deleting the text and changing the field to numeric allowed for time-based computation. Third, Acquisition_Cost had currency symbols - "\$12,724.00", demanding cleaning and numeric translation for accurate cost analysis.

To ensure consistency throughout the dataset, the original fields (Duration, Acquisition_Cost) were deleted and replaced with cleaned variables (Duration_day, Acquisition_Cost_num) to facilitate reliable processing following analytical steps.

- Duplicate records

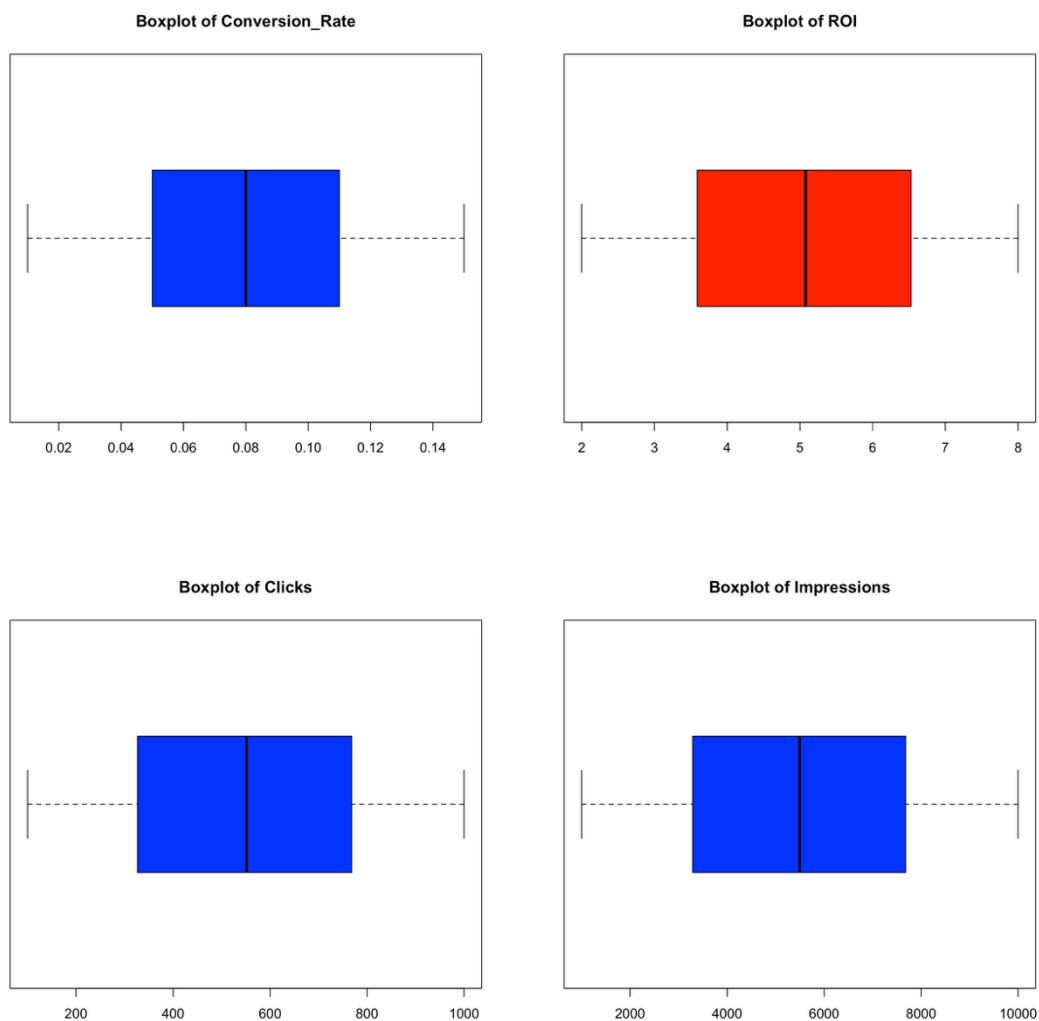
```
> ### 1. CHECK DUPLICATE RECORDS
> ## 1.1. Check fully duplicated rows (all columns the same)
> dup_flags <- duplicated(Marketing_Campaign_Dataset_A1_2_imp) | duplicated(Marketing_Campaign_Dataset_A1_2_imp, fromLast = TRUE)
> duplicates_full <- Marketing_Campaign_Dataset_A1_2_imp[dup_flags, ]
> cat("Number of fully duplicated rows:", sum(dup_flags), "\n")
Number of fully duplicated rows: 0
>
```

Figure A: R-script for duplicate checking

This dataset appears to have no duplicated rows, hence, no row is affected.

2.2. Outlier Detection and Management

To discover outliers, the boxplot approach was used because it gives a simple visual description of a variable's central tendency and dispersion while allowing for obvious identification of extreme occurrences. This approach is extensively used for outlier detection since it employs the Interquartile Range (IQR) rule to highlight data points that are not part of the main distribution.



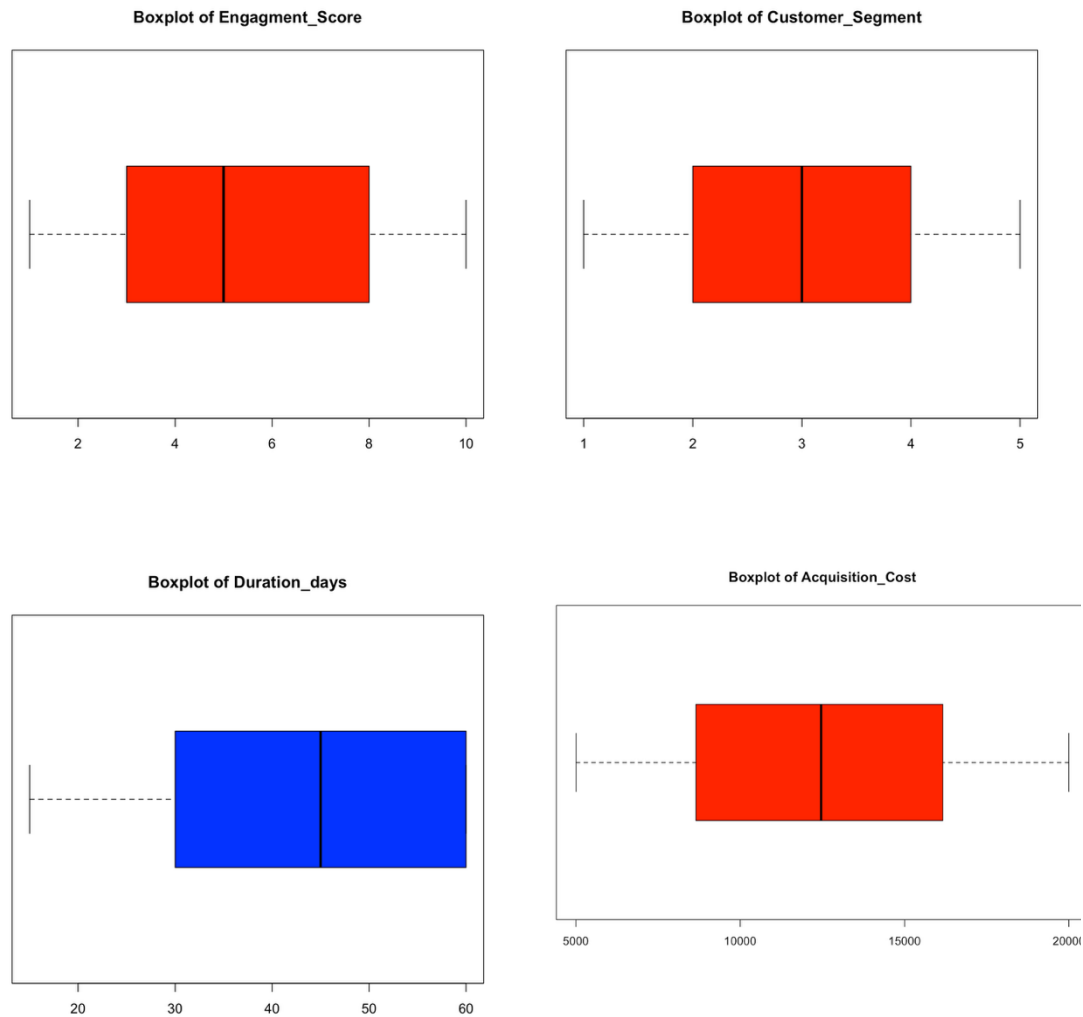


Figure B: Boxplot of numeric data

Exploratory examination revealed that there were no statistical outliers in the sample (Figure B). Skewness and kurtosis modifications are normally employed exclusively to address distributional distortions produced by outliers, therefore no such changes were necessary. The variables follow natural distribution patterns found in marketing datasets (Clicks, Impressions, ROI), and their intrinsic skewness does not jeopardize analytical integrity, interpretability, or validity. Skewness and kurtosis were examined, but not utilized as criteria for data alteration or transformation.

2. 3. Handling Missing Values

Before imputing missing values, we identify the number of missing values and the data missingness type to identify the most suitable imputation method.

Step 1: Identify number of missing values

```
> ##DATA MISSING CHECKING
> colSums(is.na(Marketing_Campaign_Dataset_A1_2_))
```

Company	Campaign_Type	Target_Audience	Channel_Used	Conversion_Rate	ROI
12	22	30	33	22	18
Location	Language	Clicks	Impressions	Engagement_Score	Customer_Segment
24	19	23	25	13	20
Gender	Age_Group	Duration_days	Acquisition_Cost_num		
30	30	22	28		

```
> sum(is.na(Marketing_Campaign_Dataset_A1_2_))
[1] 371
> na_percent <- colSums(is.na(Marketing_Campaign_Dataset_A1_2_)) / nrow(Marketing_Campaign_Dataset_A1_2_) * 100
> na_percent
```

Company	Campaign_Type	Target_Audience	Channel_Used	Conversion_Rate	ROI
0.3952569	0.7246377	0.9881423	1.0869565	0.7246377	0.5928854
Location	Language	Clicks	Impressions	Engagement_Score	Customer_Segment
0.7905138	0.6258235	0.7575758	0.8234519	0.4281950	0.6587615
Gender	Age_Group	Duration_days	Acquisition_Cost_num		
0.9881423	0.9881423	0.7246377	0.9222661		

Figure C: Data Missing Value Checking

The dataset contains 371 missing values, with individual variables ranging from 0.39% to 1.08% (Figure C), demonstrating a low degree of missingness, which facilitates the application of multiple imputation methods without significant bias. To verify methodological correctness, the missingness mechanism was further investigated to determine whether the data were MCAR (missingness depending only on seen values) or MAR (missingness unrelated to unobserved values) (Rubin 1976). Proper categorization is critical because interpreting MAR data as MCAR might mask key patterns, while treating MCAR data as MAR introduces needless modeling variance and decreases analytical dependability (Graham 2009; Gomer B & Yuan KH 2023; Gomer B 2019; Lee KJ et al. 2023). Certain imputation approaches, such as mode imputation, also rely on MCAR requirements to minimize systematic bias (van Buuren 2008).

Step 2: Data Missingness Accessing

Logistic Regression was implemented to examine the randomness of missingness (Song et al. 2013)

```

+
+   cat("\n==== LOGISTIC REGRESSION RESULTS (ENTIRE) =====\n")
+   # print(logit_results)
+
+   # Filter predictor with p < 0.05
+   significant_missing <- subset(logit_results,
+                                term != "(Intercept)" & p.value < 0.05)
+
+   cat("\n==== PREDICTORS p < 0.05 (MAR SUGGESTION) =====\n")
+   if (nrow(significant_missing) == 0) {
+     cat(">>> No predictor has p < 0.05 → missing near MCAR.\n")
+   } else {
+     # print(significant_missing)
+   }
+ }

===== LOGISTIC REGRESSION RESULTS (ENTIRE) =====

===== PREDICTORS p < 0.05 (MAR SUGGESTION) =====
>>> No predictor has p < 0.05 → missing near MCAR.

```

Figure D: Logistic Regression Results

The fraction of missing values varies between 0.4% and 1.1% across variables. To investigate the missingness mechanism, logistic regression models were fitted with a missingness indicator (1 = missing, 0 = observed) as the dependent variable and categorical predictors.

No predictors revealed p-values < 0.05 , showing no significant variations in missingness between categories. Thus, it provides insufficient backing for MAR, implying that the missing data mechanism is closer to MCAR. Given the low missingness rate and MCAR-consistent pattern, basic imputations (mean/median for numeric variables and mode for categorical variables) were used to maintain sample size without significantly altering distributions or descriptive statistics.

Step 3: Predictive Mean Matching

Single imputation (SI) is insufficient because it cannot reflect the uncertainty of missing data, but multiple imputation (MI) does. Among MI approaches, Predictive Mean Matching (PMM) is preferred because, unlike regression imputation, it preserves the original distribution of quantitative variables and maintains variance without shrinkage

bias. PMM also works best with near-MCAR patterns and minimal missingness, and it maintains imputed values realistic by relying on observed data, making it the most appropriate and balanced solution for this dataset.

However, even after applying PMM, some N/A values persisted since "N/A" was interpreted as a character string, necessitating a second imputation layer. Thus, mode imputation was used for categorical data, replacing missing values with the modal category (Memon SMz et al. 2023), whereas median-based mean imputation was utilized for numeric variables (Jadhav A et al. 2019). Although mean/median imputation is known to introduce bias (Graham JW et al. 1997; 1996; 1994), studies show that its limitations are negligible when missingness is less than 10% (Raymond MR 1986; Tsikriktsis N 2005), which is true for this dataset.

Step 4: Re-check

```
> ## Check remaining missing values
> sapply(df, function(x) sum(is.na(x)))
```

Company	Campaign_Type	Target_Audience	Channel_Used	Conversion_Rate	ROI
0	0	0	0	0	0
Location	Language	Clicks	Impressions	Engagement_Score	Customer_Segment
0	0	0	0	0	0
Gender	Age_Group	Duration_days	Acquisition_Cost_num		
0	0	0	0		

```
>
> ## 2.3.3. Check if all missing values are gone
> sapply(df, function(x) sum(is.na(x)))
```

Company	Campaign_Type	Target_Audience	Channel_Used	Conversion_Rate	ROI
0	0	0	0	0	0
Location	Language	Clicks	Impressions	Engagement_Score	Customer_Segment
0	0	0	0	0	0
Gender	Age_Group	Duration_days	Acquisition_Cost_num		
0	0	0	0		

```
>
```

Figure E: Missing Data Checking

After re-checking missing data, there are no missing values, indicating the high quality data for processing.

3. Data Visualisation

3.1. Variables selection

Conversion rates is a key sign of campaign effectiveness, providing businesses with clear visibility into performance and allowing for more efficient strategy and resource decisions (Babatunde SO 2024). Similarly, ROI is a simple indicator of financial effectiveness that compares marketing spend to revenue generated, revealing the value returned on each investment (Swetha N et al. 2024). In this research, conversion rate and ROI were initially chosen to evaluate performance and identify possible consumer categories in Scenarios 1 and 2. As Scenario 2 indicated different language-driven patterns, the variables used in Scenarios 3-5 were chosen to confirm and expand on these findings through broader behavioral and segment-level research.

3.2. Five scenarios

Scenario 1: Conversion vs ROI vs Acquisition Cost by Customer Segment

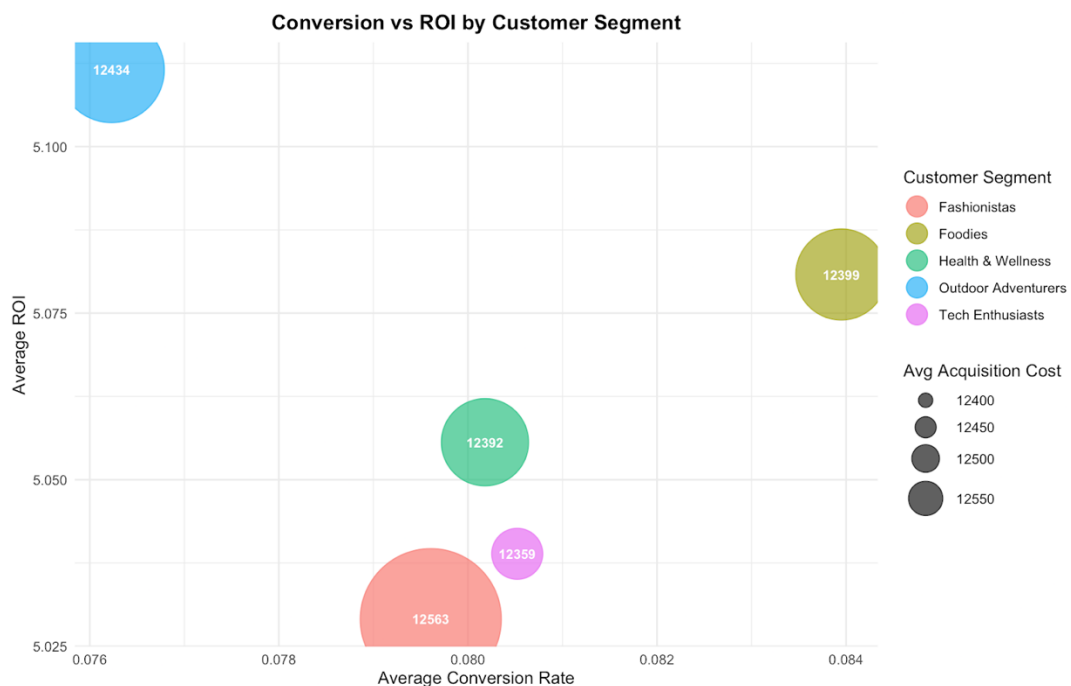


Figure 1. Bubble chart showing ROI–Conversion–Cost relationships by segment.

Figure 1 depicts five consumer categories' average conversion rate, ROI, and acquisition cost. Outdoor Adventurers have the highest conversion rate (about 0.084%), the highest ROI (around 5.11), and a relatively cheap acquisition cost. The Health & Wellness category has a conversion rate of about 0.080-0.081 and a ROI of around 5.05-5.06. Foodies have a lower conversion rate (about 0.079) and a ROI of about 5.08. Fashionistas have a lower conversion rate (almost 0.079), the lowest ROI (about 5.03), and the highest acquisition cost of any sector. Tech Enthusiasts have a conversion rate close to 0.080 and a mid-range ROI of about 5.048.

Scenario 2: Heatmap of Conversion Rate by Location and Language

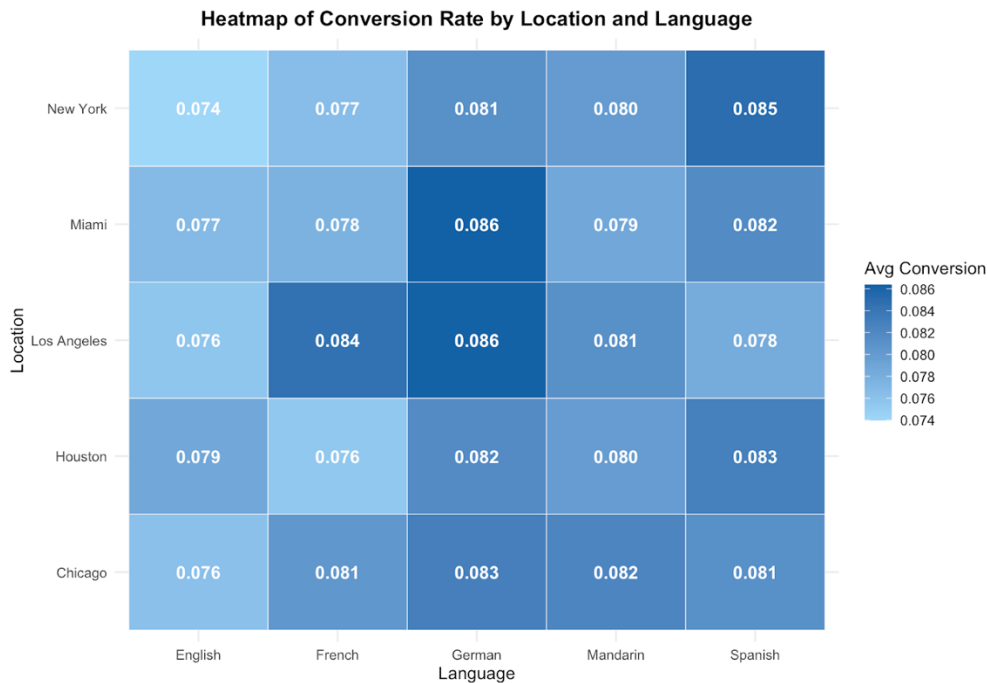


Figure 2 shows conversion rates for five places and five languages. For English, conversion values are relatively low and stable, ranging from 0.074 to 0.079 in all cities. In contrast, German had the greatest conversion rates in the sample, ranging from 0.081 to 0.086, with the highest values observed in Miami and Los Angeles (0.086). Mandarin

also has greater performance, ranging from 0.079 to 0.082, with the best scores in Chicago (0.082) and Houston (0.080).

Although English is commonly anticipated to have the best conversion rates, the evidence demonstrates that German and Mandarin regularly beat English in most places.

Scenario 3: Average Engagement by Age Group and Language

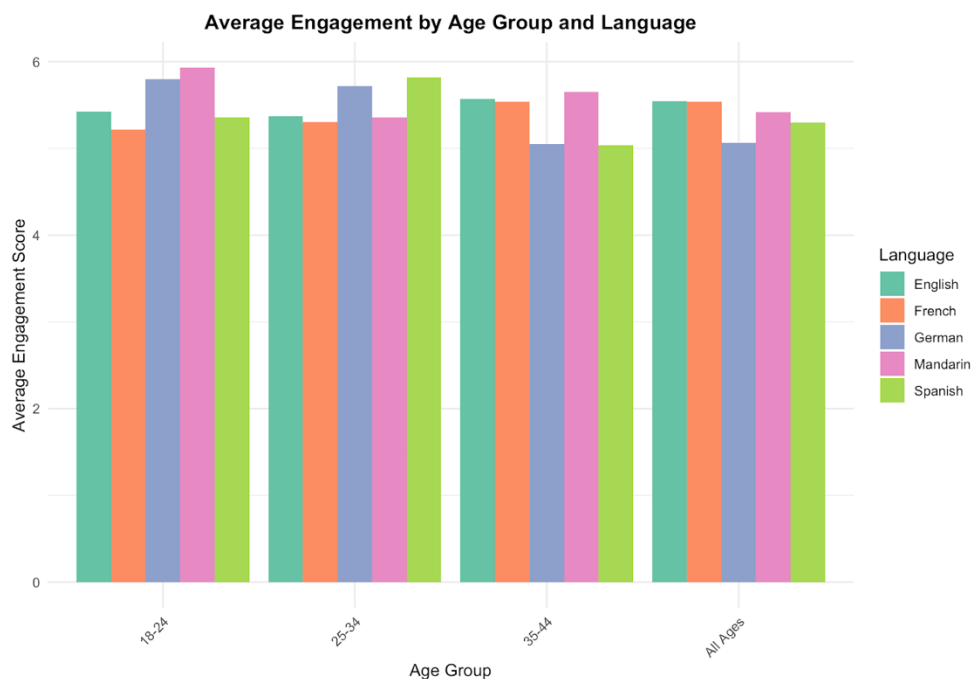


Figure 3: Age - language interaction on engagement.

The bar chart shows average engagement scores for five languages in four age categories (18-24, 25-34, 35-44, and All Ages).

Mandarin and German have the greatest engagement levels in approximately languages among people aged 18 to 24, outperforming the other three language groups significantly. Mandarin has the highest value for this age group, followed by German, making them the two most popular languages among young people.

For the remaining age categories, engagement scores are similar, with very minor variances between English, French, German, Mandarin, and Spanish. The bars in these

groups follow a regular pattern, with no significant divergence across languages as opposed to the obvious differentiation found in the 18-24 group.

Scenario 4: Conversion Rate by Language and Gender

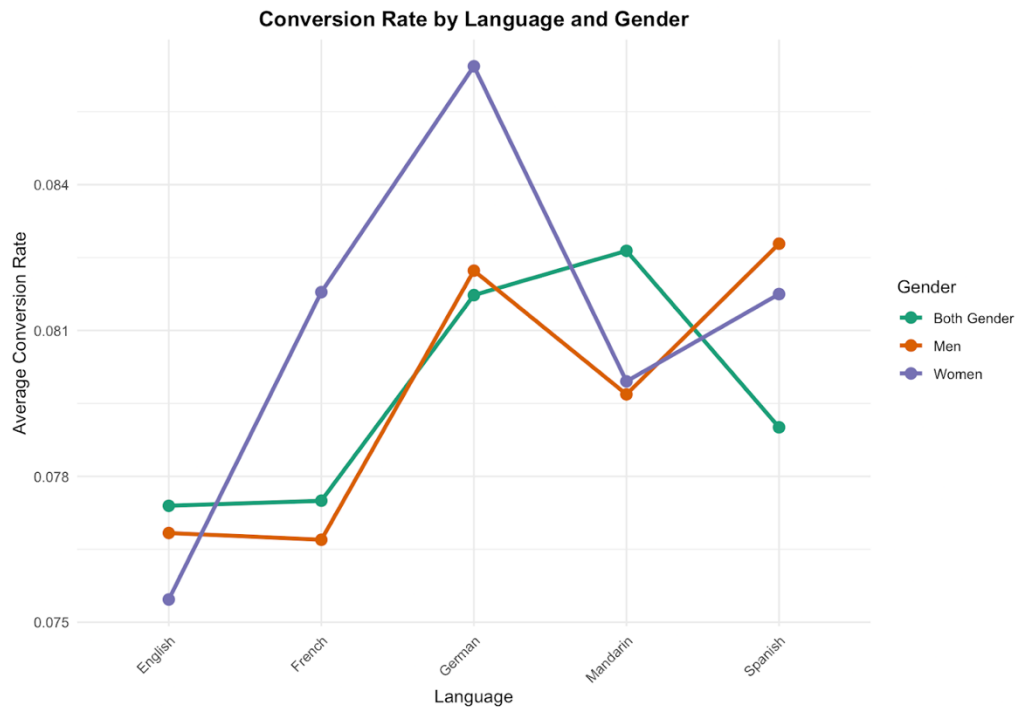


Figure 4: Line chart showing how male/female audiences convert across campaign languages.

Figure 4 shows the average conversion rates across five languages for three gender groups: males, women, and both genders together. Among all data points, German (women) has the greatest conversion rate in the whole chart, at around 0.086, greatly outperforming every other language-gender combination. Mandarin (both genders) has the second-highest total rating, at around 0.0825.

In contrast, English continually has the lowest conversion rates across gender groupings, with men, women, and both genders all clustering at the bottom of the table compared to the other languages, hovering below 0.078.

Scenario 5: Conversions and Engagement by Customer Segment and Language

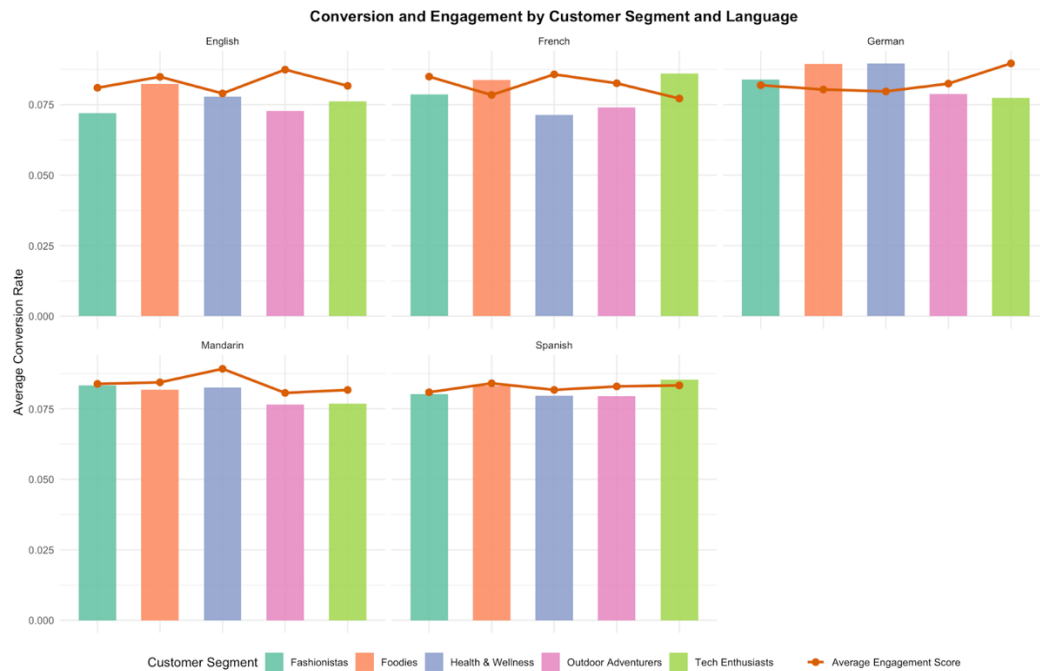


Figure 5: Conversion and Engagement rate by Customer Segment and Language

Figure 5 shows the conversion rates and engagement scores for five categories of customers in five languages. In all language panels, conversion rates for the segments range between 0.073 and 0.078, with only minor variances amongst Fashionistas, Foodies, Health & Wellness, Outdoor Adventurers, and Tech Enthusiasts. The line varies somewhat within each language but remains near the same level in English, French, German, Mandarin, and Spanish.

Across all panels, conversion and engagement numbers show little numerical fluctuation, with no significant variances between consumer segments within any language category.

4. Findings and Discussion

Scenario 1:

The results demonstrate significant performance disparities across parts. Outdoor Adventurers outperform with the greatest conversion (0.084%), ROI (5.11), and low acquisition cost, consistent with studies suggesting experience-oriented customers

respond strongly to immersive, action-driven creative (Olney TJ et al. 1991; Haans H et al. 2013; Rosener W 2023). Fashionistas, on the other hand, have the lowest conversion (0.079), ROI (5.03), and acquisition cost—meaning more spend is required per acquired customer—confirming findings that fashion operates in saturated, high-CPM markets where creative fatigue reduces effectiveness (Lambrecht & Tucker 2013). This is supported by McKinsey (2023), which shows acquisition costs increasing from \$9 in 2013 to \$29 in 2022, indicating a fall in third-party targeting and the need for companies to diversify platforms and improve creative quality.

Foodies have a small conversion rate (0.079) but a larger ROI (5.08), which is consistent with research indicating that food purchases are influenced by social proof and browsing behavior (Begho T & Liu S 2024; Hilverda F et al. 2018; Hasanein AM & Montaser NM 2025). Tech Enthusiasts convert slowly (0.080) but maintain a constant ROI (5.048), consistent with data that tech users perform lengthy research and make higher-value purchases (Strebel J et al. 2004; Peterson RA & Merino MC 2003; Grant R et al. 2007; Mogaji E & Jain V 2024). These findings highlight behavioral differences across experiential, fashion-driven, food-motivated, and information-driven consumer groups.

Strategically, Ogilvy should increase investment for Outdoor Adventurers (10–15% test uplift), optimise Fashionista costs via short-form content, retargeting, reduced cold spend (10–20%), and A/B tested informative creatives; apply an engagement-first plus retargeting approach for Foodies; and provide deeper informational content (explainers, testimonials, case studies) for Tech Enthusiasts before adjusting budget levels.

Scenario 2:

German-language advertising consistently had the greatest conversion rates across virtually all locales, particularly in Miami and Los Angeles, indicating substantial localization effects and possibly stronger buy intent among German-speaking or culturally related consumers. In contrast, English-language advertisements, although being the norm in US markets, had the lowest conversion rates, consistent with studies indicating that localized content frequently outperforms English-only marketing even in English-dominant areas. Spanish and Mandarin scored relatively well, demonstrating that

multilingual initiatives offer clearer engagement and conversion benefits than wide English ads. Markets such as New York and Chicago have lower conversion rates across most languages, indicating increased competition noise, a poorer message-market fit, or creative mismatch.

These findings are consistent with multilingual advertising literature, which shows that audiences respond more strongly to languages perceived as culturally aligned, visually structured, or emotionally resonant (Clarissa LA 2025; Rigby BD & Lee R 2024; Musonda N & Siame P 2025; Turobova OO & Akmalxonov SFA 2024). German and Mandarin creatives frequently use high-context cultural clues and organized messaging to increase persuasion and clarity (Irizarry QA et al. 2023).

Economic position may potentially impact these patterns, since financial capability influences purchasing behavior (Kumar V et al. 2014). U.S. Asian families, with Chinese Americans accounting for the biggest grouping, report a median income of \$104,646, which is almost 30% more than the U.S. median of \$74,755 (USAFacts n.d.), but German immigrants also outearn the U.S. median. (Appendix C-D). These wealth disparities may explain why English-language advertisements underperform while dominating the market.

Overall, the findings underline the need of tailoring marketing wording to individual target segments. For Ogilvy, this entails prioritizing multilingual content testing in high-performing regions, improving localization in underperforming ones, and minimizing reliance on English-only messaging for diverse or multicultural audiences.

Scenario 3:

This is consistent with research showing that Generation Z is the most open to intercultural, cross-cultural, and transnational media aesthetics (Yusuf A 2023; Koohestani AS 2024). Users aged 25 and up, on the other hand, have more stable and less exploratory content choices (Birk MV et al. 2017; Bian J et al. 2014; Karumur RP et al. 2018), resulting in a smaller engagement variance between languages. The dataset reflects these trends: Gen Z is the most sensitive to language-based creative variations, whereas older cohorts participate more consistently across languages.

The scenario 2 and 3 indicates that Germany and Mandarin Gen Z could be the main drivers for the high conversion rate among major cities in the U.S. Therefore, it is possible that this generation, which has the greatest conversion rates, is drawn to advertising in languages that are regarded to be culturally connected, visually organized, or emotionally resonant.

Hence, one key insight for Ogilvy while expanding the operation globally is to focus on localizing key marketing campaign messages targeting Gen Z (18-24) to maximize the profit.

Therefore, it is essential to analyse the customer preferences and customer personas detailly to select the proper marketing strategy targeting at these groups, this papers conduct Scenario 4: Conversion Rate by Language and Gender to investigate the gender variables in this cohort, and Scenario 5: Conversion and Engagement by Customer Segment and Language to investigate their preference field.

Scenario 4:

The findings indicate that German-language commercials for women have the greatest conversion rate in the sample (0.086), followed by Mandarin for both genders (0.0825). English consistently has the lowest conversion rate among men, women, and both genders (<0.078). This pattern is consistent with data demonstrating that German girls, particularly Gen Z (18-24), who are highly engaged on social media (as described in Scenario 3), are Ogilvy's most profitable client group, with a high conversion rate (as mentioned in Scenario 1).

Meanwhile, Mandarin-generated commercials have a relatively good conversion rate across both genders. This demonstrates a favorable association between Mandarin commercials and both gender conversion rates, emphasizing the need for neutral advertisements - goods for both genders such as food or essentials. In contrast, English-

language commercials are often targeted broadly, resulting in poorer audience qualification and conversion efficiency (Lambrecht & Tucker 2013).

Based on these findings, Ogilvy should emphasize German-language advertisements for female consumers while expanding Mandarin-language creative to larger target groups when suitable. English should not be utilized as the default language for performance-driven goals; rather, it should be combined with more specific targeting or used for upper-funnel exposure. Testing more localized language variations and altering gender-based creative customization can help to increase conversion efficiency.

Scenario 5:

Across languages, conversion and engagement remain generally stable, but Mandarin and German reveal clearer segment-level contrasts.

In German, Foodies and Health & Wellness had the best conversion rates despite only moderate engagement, suggesting that although message effectively drives action, content may not be compelling enough to sustain increased top-funnel involvement. According to Li S (2025), low engagement can restrict long-term scalability and repeat conversion potential; hence, a good long-term strategy must be developed to prevent over-optimism about the growing numbers in this area. Conversely, German Tech Enthusiasts have the highest engagement but the lowest conversion rate, which is consistent with findings suggesting 34% of German customers interact significantly yet require lengthy study and trial before committing to technology items (Statista 2025). This implies the necessity for experiential or offline touchpoints, such as demos, trials, or more in-depth instructional content, to take them from curiosity to conversion.

Fashionistas, Foodies, and Health & Wellness all have good conversion rates in Mandarin, but only Health & Wellness has high engagement, indicating that Chinese adolescents choose health care (Statista 2024). For Fashionistas and Foodies, where conversion is strong but engagement is low, Ogilvy should determine whether existing Mandarin creatives lack aspirational components or sensory depth. Structured A/B testing

on visual design, influencer fit, and messaging tone might help uncover changes that increase engagement without reducing conversion. (Kotwal, R 2025)

Overall, these conversions - engagement mismatches indicate that while some segments already convert efficiently, strengthening emotional engagement presents a clear opportunity to enhance sustainability and scaling potential across Mandarin and German campaigns.

5. Conclusions

This research showed how Descriptive Analytics can be utilized to transform raw global campaign data into useful insights for Ogilvy International. The study revealed which consumer categories, languages, age groups, and genders produce the highest conversion and ROI, as well as where engagement-conversion gaps suggest untapped potential, thanks to thorough data pretreatment and five tailored visual scenarios. As a Marketing Campaign Manager, the essential takeaway is that performance is not uniform: it is influenced by the combination of segment identification, localization strategy, and audience behaviour. Prioritizing high-performing German and Mandarin Gen Z cohorts, while redesigning approaches for saturated fashion audiences and research-heavy tech consumers, gives Ogilvy a clear, evidence-based path to increase efficiency, support better resource allocation, and strengthen the long-term effectiveness of its international marketing campaigns.

6. Recommendations

Recommendation 1: Address fashion saturation and redesign approach for Tech Enthusiasts

Insight (from Scenario 1): Fashionistas show poor conversion and ROI with high acquisition costs, indicating saturation, whereas Tech Enthusiasts have equally modest conversion but consistent ROI, driven more by brand and offline experience than digital exposure alone.

Fashion is one of the most crowded digital industries, with constantly escalating client acquisition prices. Ogilvy should prioritize cost management and optimization for the Fashionistas sector, reviewing and testing ad placement, format, and frequency to discover situations where acquisition cost and conversion are less limited, rather than just boosting spend in congested channels.

In parallel, Tech Enthusiasts should not be managed as a saturation problem, but as a consideration and experience problem: strategies should emphasise brand credibility, detailed information, and hybrid offline-online experiences, in line with evidence that a large share of shoppers conduct extensive research and frequently prefer to see or try products—particularly higher-involvement or premium items—before purchasing. The remaining segments (Outdoor Adventurers, Foodies, Health & Wellness) perform satisfactorily, but Ogilvy should position them within a long-term portfolio strategy—maintaining investment while gradually testing creative, channel, and localization refinements to maintain current efficiency as acquisition costs rise globally.

Recommendation 2: Localised Gen Z targeting in German and Mandarin segments

Insight (from Scenarios 2–5): German (female) and Mandarin (both genders) Gen Z audiences in the United States have the highest conversion rates, particularly among Foodies and Health & Wellness, whereas Tech Enthusiasts have strong engagement but poor conversion.

As Marketing Campaign Manager, I recommend prioritizing localized campaigns for German-speaking Gen Z women and Mandarin-speaking Gen Z (both genders) in the United States, utilizing German and Mandarin language, visuals, and cultural cues to align with Gen Z's preference for culturally relevant, identity-reflective content. (EQHO 2024; Koohestani AS 2024; Kalakrit, undated)

Within these cohorts, Ogilvy should create richer, more emotionally and sensorially engaging creatives for Foodies and Health & Wellness to increase engagement, which is currently lower than their strong conversion, and thus protect long-term conversion

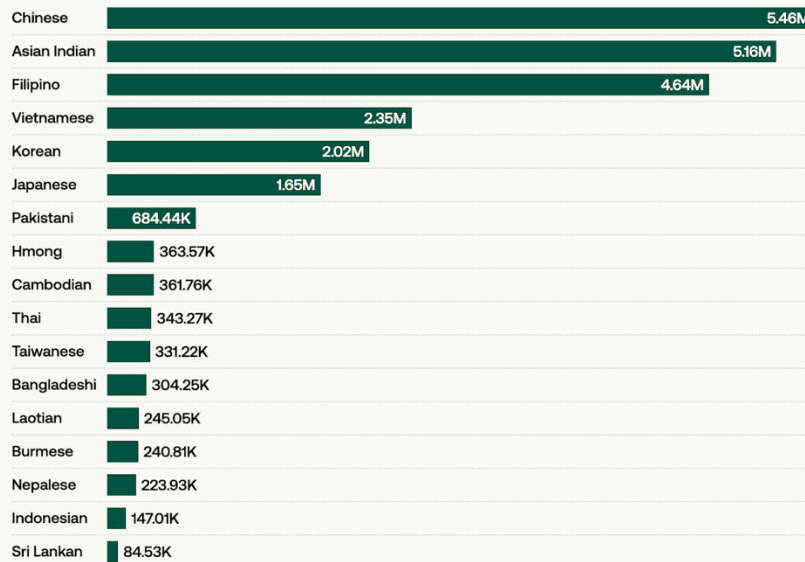
performance rather than relying solely on short-term efficiency. For Tech Enthusiasts, where engagement is high but conversion is low, Ogilvy should shift to research- and experience-led journeys, reflecting evidence that many consumers—particularly in the tech and premium categories—do extensive research and prefer to try products or evaluate brands before buying. (SellersCommerce 2025; Basu R, Sondhi N 2021).

Given the observed gender gap in engagement, German campaigns in these locations should prioritise female-focused propositions, while Mandarin campaigns should emphasise essential, neutral products suitable for both genders, directly aligning language, segment and gender with the highest observed conversion clusters in the data.

Appendices

Chinese Americans are the most common Asian American ethnic group.

Number of people by Asian ethnicity, 2023



Figures include those that identify as one ethnicity in combination with another. Due to mixed identities, these figures do not sum exactly to the total number of Asian Americans in the US.

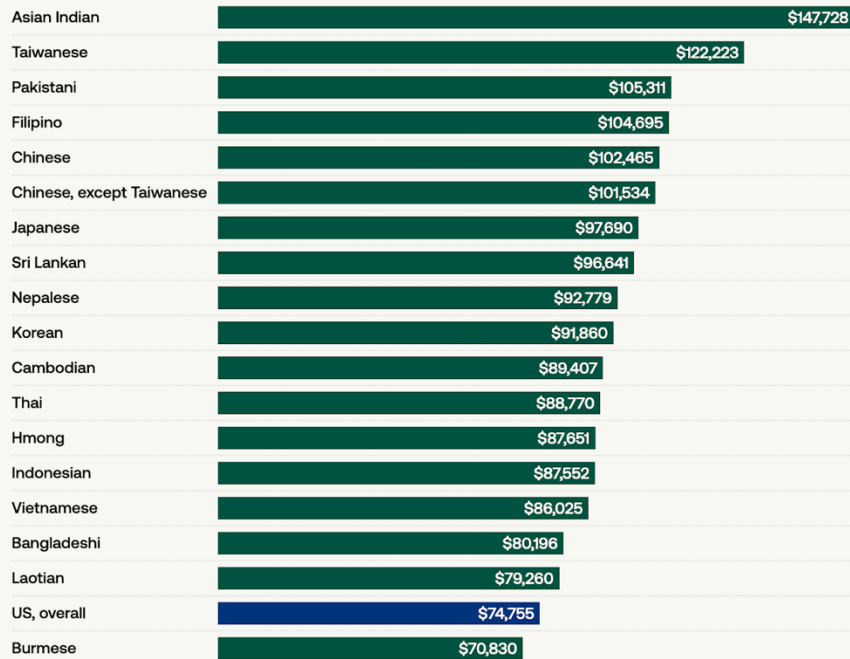
Source: [US Census Bureau](#) • [Get the data](#) • [Embed](#) • [Download image](#) • [Download SVG](#)

Appendix C: Chinese Americans are the most common Asian American ethnic group

Source: <https://usafacts.org/articles/the-diverse-demographics-of-asian-americans/>

Indian Americans bring in nearly twice the income of the average American.

Median household income by Asian ethnicity, 2022



Figures include those that identify as one ethnicity in combination with another.

Source: [Census Bureau](#) • [Get the data](#) • [Embed](#) • [Download image](#) • [Download SVG](#)

Appendix D: Indian Americans bring in nearly twice the income of the average American. Source: <https://usafacts.org/articles/the-diverse-demographics-of-asian-americans/>

Median Earnings Comparison

AMERICANS

Tragic

\$42,742

MEDIAN EARNINGS

0.6 / 100

METRIC RATING

268th / 347

METRIC RANK

IMMIGRANTS FROM GERMANY

Excellent

\$47,566

MEDIAN EARNINGS

83.1 / 100

METRIC RATING

152nd / 347

METRIC RANK

American vs Immigrants from Germany Median Earnings Correlation Chart

“ The statistical analysis conducted on geographies consisting of 581,345,964 people shows a slight negative correlation between the proportion of Americans and median earnings in the United States with a correlation coefficient (R) of -0.096 and weighted average of \$42,742. Similarly, the statistical analysis conducted on geographies consisting of 463,817,606 people shows a mild negative correlation between the proportion of Immigrants from Germany and median earnings in the United States with a correlation coefficient (R) of -0.319 and weighted average of \$47,566, a difference of 11.3%.

Appendix E: Median Earnings Comparisons of American vs Immigrants from Germany Median Earnings. Source: <https://zipatlas.com/us/income-comparison/americans-vs-immigrants-from-germany-median-earnings.htm>

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